

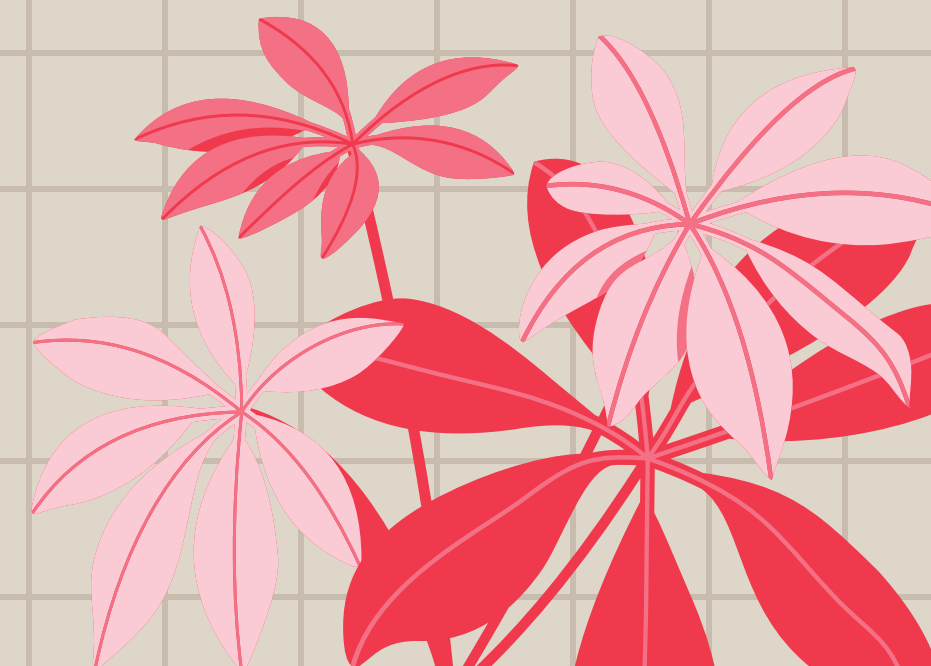
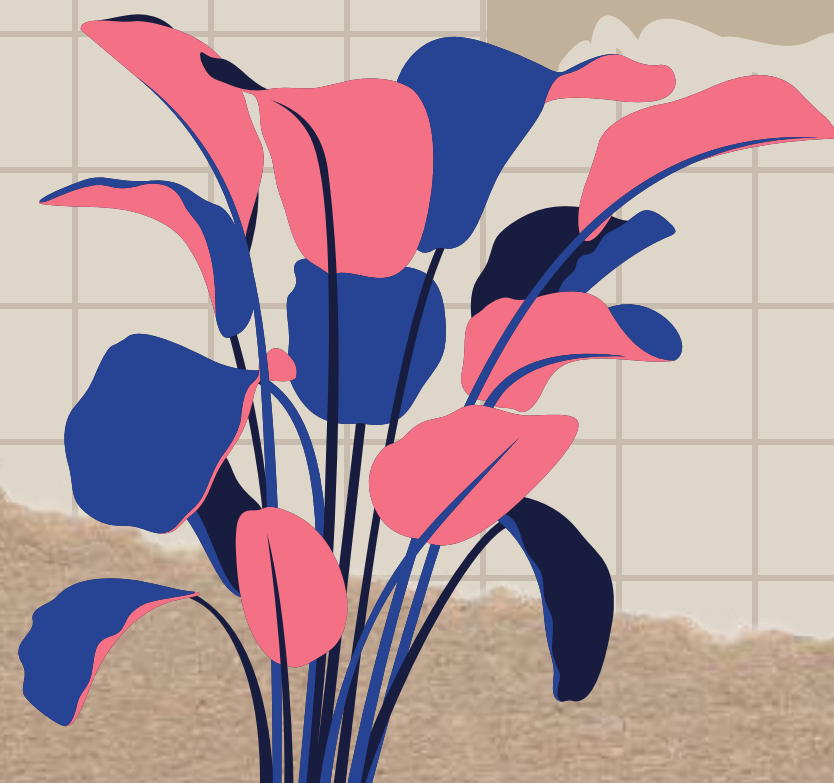
**IEEE Build-a-thon
2026**

Team 101

Presenting...

VYSL

a vision for the future



MEMBERS

Tyler Grogg
Shreyans Bhuyan
Catherine Satya
Daniel Kang





THESIS

Our project, Vyse, is a prototype designed to detect rapid and repetitive patterns of bright light that can trigger epileptic seizures. It provides real-time alerts to the user to help them with their safety, and improves independence in everyday lives.



MOTIVATION

SOCIAL GOOD

**HELPING THOSE WITH
DISABILITIES**



Our project aims to detect rapidly changing light levels to alert the wearer know if they're currently in an seizure-inducing environment by sending a gentle vibration to their head.

Even when lights flash faster than is perceptible, it can still be overwhelming and trigger photosensitive epilepsy.

<https://www.epilepsy.com/what-is-epilepsy/seizure-triggers/photosensitivity>

MOTIVATION

cont.

We know that it can be difficult to tell at times whether a room's lights are flashing or not. If you've ever taken a video of a light source and seen it flickering, you know this phenomenon. Being that we can't see it with the naked eye, it creates a dangerous environment for someone susceptible to flashing lights.

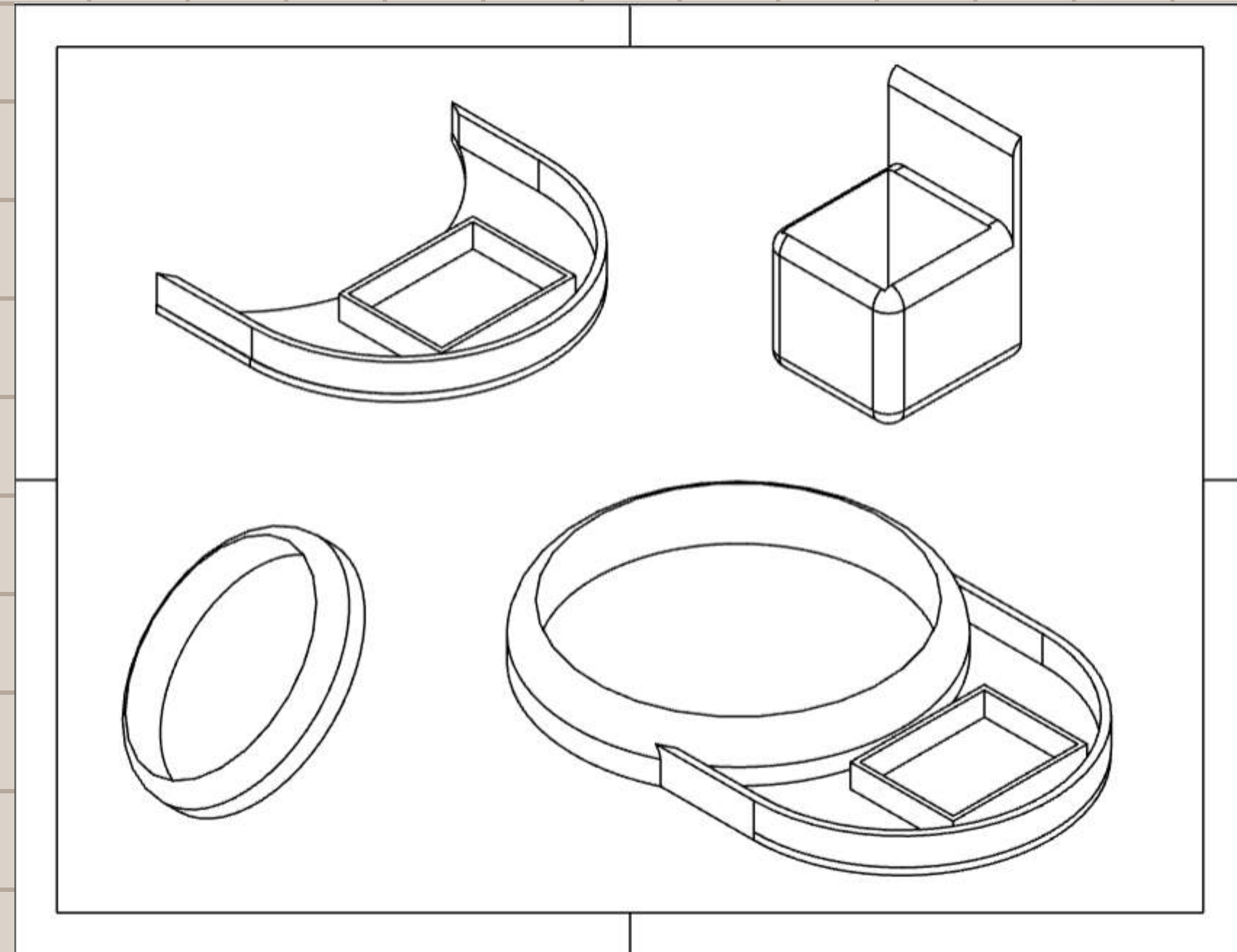


GOALS

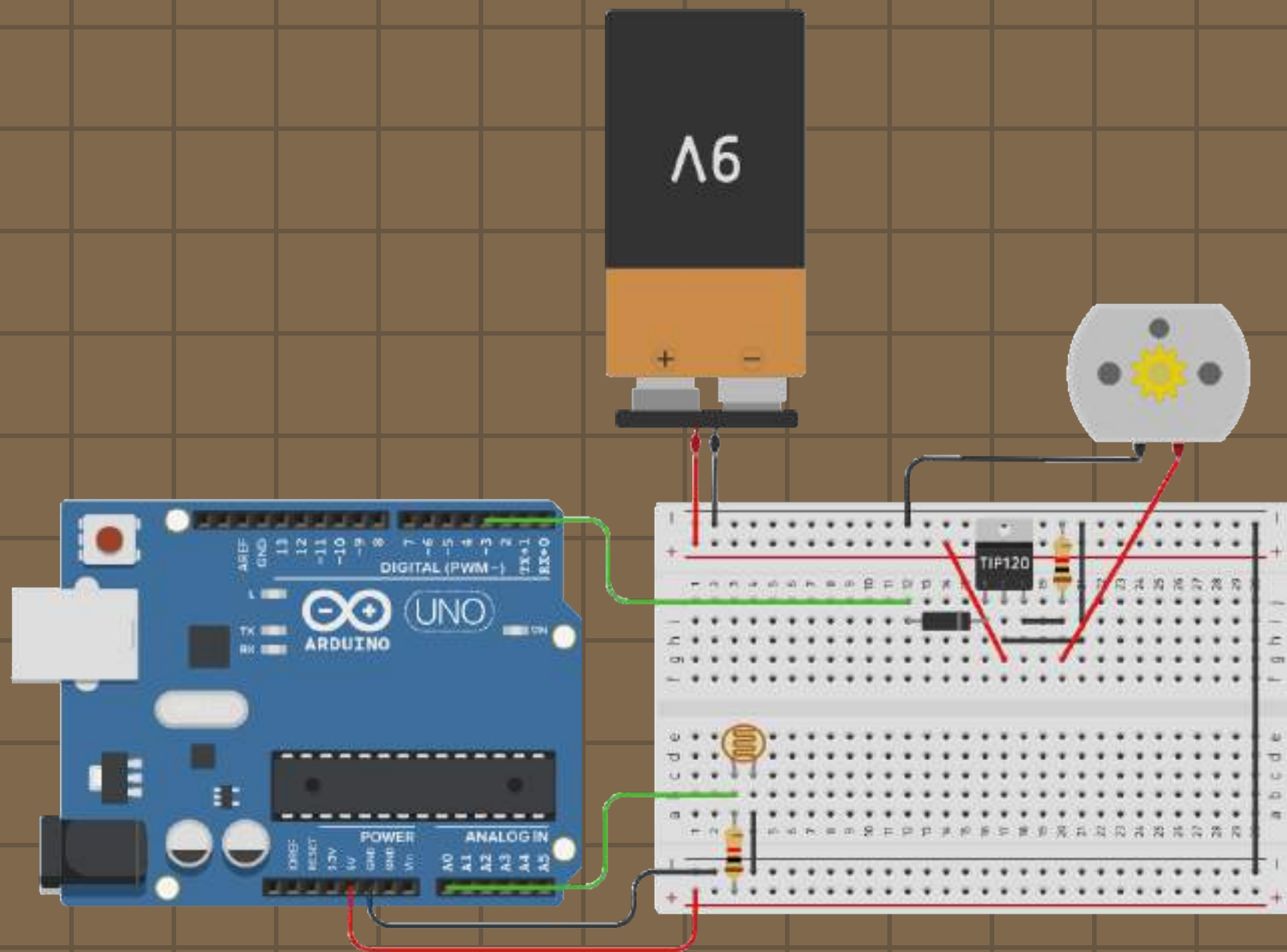
- Help those with epilepsy avoid potential danger
- Create a prototype to demonstrate feasibility of future products
- Ensure accessibility to those who are in need



VISOR SCHEMATIC



CIRCUIT



Key Components

- diode to prevent voltage spiking
- transistor to connect battery and motor
- photoresistor to detect changes in light

COMPONENTS / BUDGET

Open Arduino – \$30

Arduino Cable – \$5

3–6V Motor – \$6

Photoresistor – \$3

9V Battery – \$10

Transistor – \$6

Wires – \$4

Diode – \$3

Battery Connector – \$2

Total = \$69



ARDUINO PSEUDOCODE

- instantiate global variables (array size, expiration time, vibration length)
- each loop, check the current through the photoresistor
 - if above or below the last value by a predetermined amount (set based on the environment), add the value and timestamp to array
- see if the array has more than 10 values that haven't timed out
 - if so, send a notification via vibration of the motor



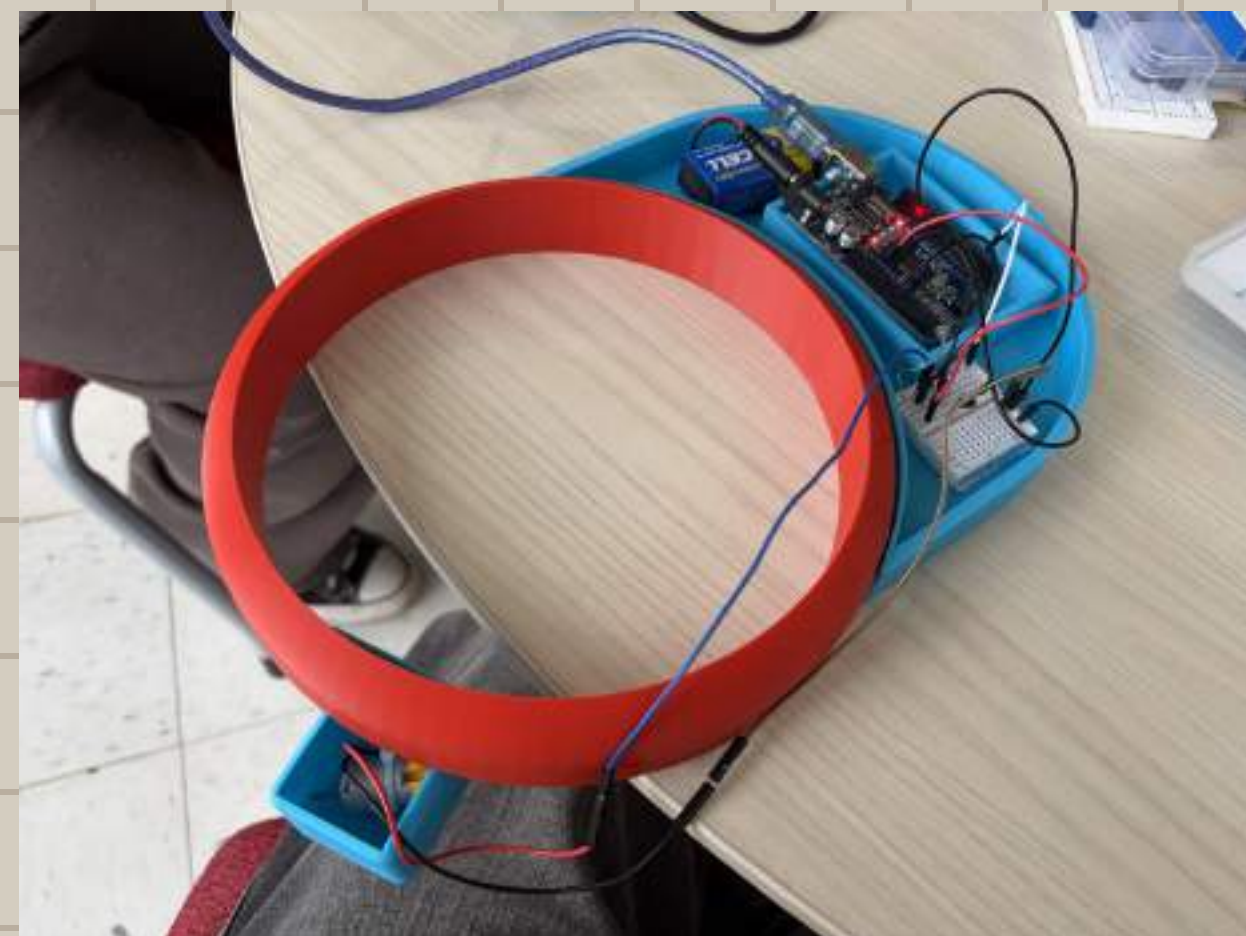
ARDUINO CODE

```
1 // C++ code
2 //
3 const int ARRAY_SIZE = 10;
4 const unsigned long EXPIRATION_TIME = 500; // Time in milliseconds
5
6 const unsigned long MOTOR_DURATION = 1000;
7 bool motorRunning = false;
8 unsigned long motorStartTime = 0;
9
10
11 double currThreshold = .05; //mA
12
13 double prev_current = 0;
14 double current = 0;
15 double voltage1 = 0;
16
17 double dataArray[ARRAY_SIZE]; //document last x currents
18 int index = 0;
19 unsigned long entryTimes[ARRAY_SIZE];
20 unsigned long lastFillTime = 0;
21
22 void setup()
23 {
24     Serial.begin(9600);
25
26     int sensorValue1 = analogRead(A0);
27     double voltage1 = sensorValue1 * (5.0 / 1023.0);
28     double current = 1000.0*((5.0-voltage1)/1000.0); //resistor value
29     prev_current = current; //set up first loop as prev_current = cur
30
31     for (int i = 0; i < ARRAY_SIZE; i++) { //fill with zeroes
32         dataArray[i] = 0;
33         entryTimes[i] = 0;
34     }
35     pinMode(3,OUTPUT);
36 }
37
```

```
37
38 void loop()
39 {
40     unsigned long currentTime = millis();
41
42     int sensorValue1 = analogRead(A0);
43     double voltage1 = sensorValue1 * (5.0 / 1023.0);
44     double current = 1000.0*((5.0-voltage1)/1000.0); //resistor value = 1k
45
46
47     //Serial.print("The current is: ");
48     //Serial.println(current); //in mA
49
50     if (1(current <= prev_current+currThreshold && current >= prev_current-currThr
51
52         Serial.println("Bad lights detected.");
53         lastFillTime = currentTime;
54
55         dataArray[index] = 1;
56         entryTimes[index] = currentTime;
57
58         Serial.print("Added value to index [");
59         Serial.print(index);
60         Serial.println("]");
61
62         index++;
63         if (index >= ARRAY_SIZE) {
64             index = 0;
65         }
66
67         int count = 0;
68         for (int i = 0; i < ARRAY_SIZE; i++) {
69
70             // Only check spots that aren't already empty (0)
71             if (dataArray[i] > 0.0) {
72
73
```

```
73
74
75         Serial.println("REALLY BAD lights detected. Running notification.");
76         //insert motor functions here
77         digitalWrite(3, HIGH);
78
79         motorStartTime = currentTime; // Record the "Start Time"
80         motorRunning = true; // Set the flag
81
82     }
83
84 }
85
86 }
87
88 }
89
90 }
91
92 for (int i = 0; i < ARRAY_SIZE; i++){
93     if (dataArray[i] != 0) {
94         if (currentTime - entryTimes[i] >= EXPIRATION_TIME) {
95             dataArray[i] = 0; // Clear the value
96             entryTimes[i] = 0; // Reset the timer
97
98             Serial.print("Index [");
99             Serial.print(i);
100             Serial.println("] expired and cleared.");
101         }
102     }
103 }
104
105 if (motorRunning) {
106     if (currentTime - motorStartTime >= MOTOR_DURATION) {
107         digitalWrite(3, LOW); // Turn motor/LED off
108         motorRunning = false; // Reset the flag
109         Serial.println("Motor duration complete.");
110     }
111 }
112
113 delay(10);
114 prev_current = current;
```

DEMO REEL



CHALLENGES

A challenge we faced was deciding on what we wanted to build for the project that we could actually complete within our skill level in the limited time given for the build-a-thon.

Another challenge was limiting the scope of our project. Making sure we didn't overextend ourselves and had a realistic timeline was crucial to our success.



FUTURE IMPROVEMENTS

Incorporate computer vision models to automatically detect light levels and adjust the parameters.

More photoresistors for better, 360 degree detection.

Have future designs be more discreet to encourage daily use and be suitable for any occasion.

Adding a clear glass lid on the top of the visor so components won't fall out, and the photoresistor is able to work.

Use an actual fabric visor with adjustable strap for comfort and , to be accesible for people with different head sizes.

THANK YOU!